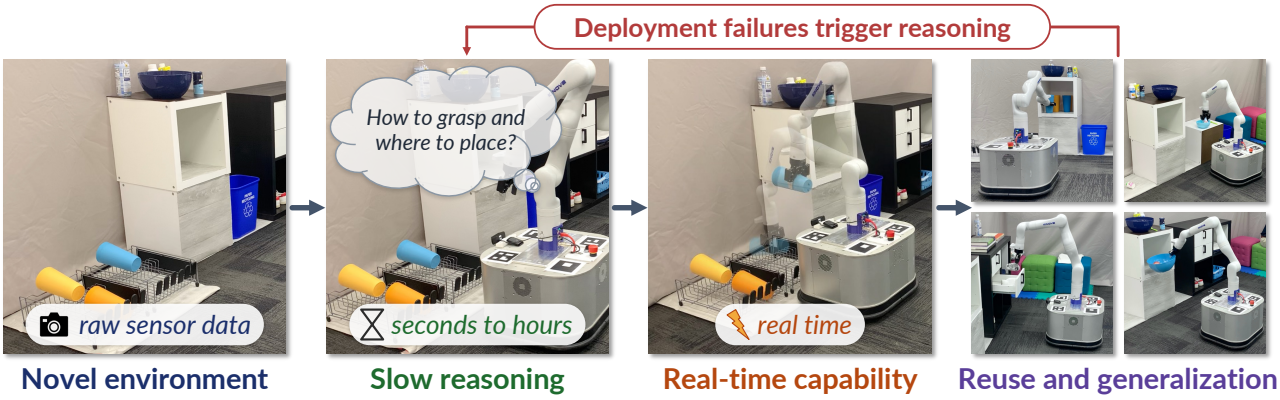


Figure 1: My research aims to enable robots to solve novel tasks through slow reasoning, distill solutions into real-time, reusable visuomotor policies, and generalize these capabilities to new tasks and environments. Deployment failures trigger further reasoning, allowing robots to continually expand what they can do.

I aim to build robots that continually expand their physical capabilities through experience: solve unfamiliar tasks, acquire reusable capabilities, and adapt to novel environments, embodiments, and users. Such robots are needed in homes, care facilities, industrial settings, and disaster zones, where they must solve novel tasks and adapt to changing needs. Achieving this vision requires both the ability to **reason** through unfamiliar tasks and the **real-time robot capabilities** for physical interaction. For instance, when helping a care recipient drink from a new mug, a robot must determine how to grasp and maneuver it around nearby obstacles while positioning it at an appropriate angle and distance for the user to drink. These constraints are interdependent: a grasp that is feasible on its own may restrict the orientation of the mug, preventing the robot from safely bringing it to the recipient. During deployment, the robot must continuously react to changes in the pose of the mug and the movements of the recipient, and if the mug slips, it should learn from that failure so it handles similar situations next time. My research aims to enable robots to solve unfamiliar tasks through slow reasoning, turn their solutions into real-time, reusable capabilities, and continually improve through experience. This creates a virtuous cycle that continually expands what robots can do (Fig. 1).

Existing approaches address different aspects of this challenge. Task and motion planning (TAMP) [3,4] and recent coding agents [5] enable robots to reason through unfamiliar tasks by interleaving symbolic planning and continuous geometric planning, or by generating programs, invoking specialized tools, testing solutions, and iteratively refining strategies. However, such reasoning can take seconds to hours, making it impractical to invoke continuously during physical interaction. In contrast, learned visuomotor policies map visual observations directly to robot actions, enabling fast responses to changing physical environments. Yet learning policies that handle the diversity of real-world interactions requires extensive data, and pretrained policies still struggle to generalize to novel environments and tasks beyond their training distributions [1,2]. Furthermore, neither paradigm systematically turns deployment failures into improvements: reasoning-based approaches rarely update their models during deployment, and learned policies require costly new data to correct their failures.

My past work has explored the key components of this reasoning-to-capability cycle. First, I developed **the first methods for learning abstract world models that enable hierarchical planning directly from partial-view point clouds** [6–8], enabling reasoning on real robots directly from raw sensory observations. Second, I developed methods that make reasoning more efficient by distilling reasoning directly into fast vision-language-action models (VLAs) that can be reused for future tasks [11,12], and by learning score functions to prioritize promising plans, improving planning efficiency and success rates [9,10]. However, robots can still fail during deployment when their world models or reusable capabilities are insufficient for the situation. Third, I developed systems that identify failures as signals for improving future capabilities [13,14].

Building on these foundations, my research agenda is to develop robots that continually expand their physical capabilities throughout deployment as they encounter novel tasks, environments, embodiments, and users. I will pursue this agenda through four complementary directions: (1) structured robot memory to maintain decision-relevant physical knowledge, (2) uncertainty-aware reasoning to make reliable decisions in unfamiliar environments, (3) generalizing capabilities across diverse embodiments by learning transferable physical knowledge, and (4) specializing generalized capabilities to individual users through a personalization layer.

1 Past Work

1.1 Enabling Reasoning from Raw Sensor Data

To reason about physical problems, robots should predict how their actions will change the environment and use these predictions to plan toward a goal, despite having only partial, noisy sensor observations. To address this challenge, I developed the first methods for learning abstract, object-centric world models to enable hierarchical planning from partial-view point clouds [6–8, 15].

A key question I addressed is how to learn a representation directly from partial-view point clouds that can capture both spatial relations (e.g., the book is in the drawer) and temporal effects of robot actions (e.g., which relations would change if the robot sweeps the trash on the table). I addressed this question by integrating spatial-temporal reasoning over a learned latent representation from sensory data shown in Fig. 2. This representation captures both geometric and symbolic effects of actions within a shared latent space, and the learned latent dynamics model can predict future latent states to enable planning. I integrated these learned world models with hierarchical planning to enable robots to solve long-horizon manipulation tasks directly from partial-view point clouds. Given a language instruction and a partial scene observation, my approach uses sampling-based planning to achieve the goal while accounting for geometric constraints and object interactions. I demonstrated this capability on a real mobile manipulator across diverse tasks, including placing multiple objects in a geometrically constrained cupboard (Fig. 2). In real-world evaluations, my approach successfully solved 86.7% of the long-horizon tasks [8], demonstrating its capability in complex environments.

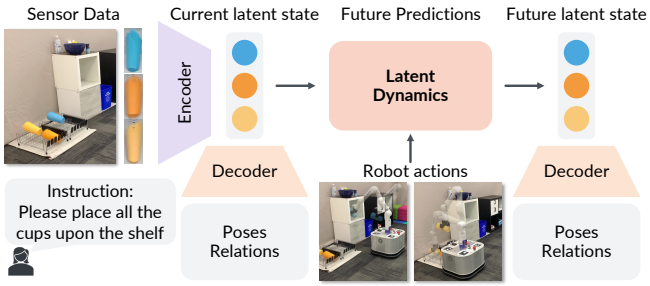


Figure 2: From partial-view point clouds, the robot learns an object-centric world model that predicts how robot actions change the environment, enabling goal-directed planning directly from raw sensory observations.

1.2 Turning Reasoning into Real-Time, Reusable Capabilities

Reasoning is powerful but expensive. Rather than repeatedly paying this cost, I have explored two complementary ways to make reasoning real-time and reusable: distilling reasoning into reusable visuomotor policies, and learning from previous experience to make future planning faster.

First, with my mentees Samrat and Liang, we explored distilling slow reasoning into real-time, reusable capabilities. In DATAFARM [11], we used TAMP to generate demonstrations offline and train fast VLAs that execute reactively on the real robot. We found that planner-generated trajectories are ineffective unless their behavioral distribution is aligned with VLA pretraining data. Fine-tuning on these aligned demonstrations raises success from 8.3% to 56.7%, approaching the oracle baseline that uses human teleoperation (61.7%). TANDEM [12] extends this approach to behaviors beyond what the planner can generate by selectively using human teleoperation. Together, these projects demonstrate how slow reasoning can be transformed into reactive policies that can be reused during real-time physical interaction.

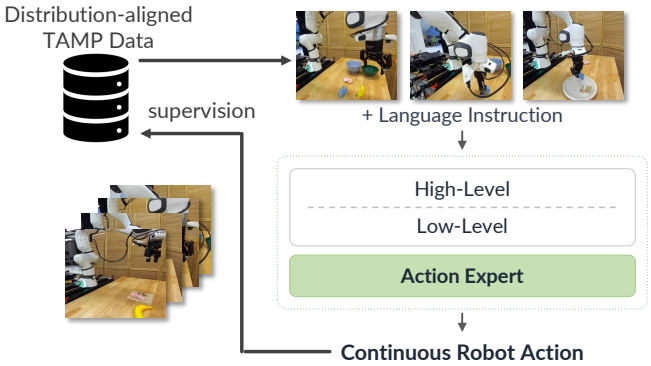


Figure 3: Distribution-aligned task and motion planning (TAMP) demonstrations provide supervision for fine-tuning downstream VLAs, transforming solutions generated through reasoning into reactive policies for real-time robot execution.

Second, my mentees Joseph, Qihang, and I developed methods that learn to accelerate reasoning itself. In SPECTRE [9], we developed a learned plan ranker that uses evidence from failed motion refinements to prioritize promising plans. In SC-BPE [10], we similarly learned to score candidate base placements jointly with plan skeletons before expensive refinement. Across eight task settings and two mobile manipulators, SC-BPE raises the average success rate from 54% to 89% and reduces the

average planning time by 50% compared with random ordering of the same candidates. In these two projects, the robot reuses learned guidance to make subsequent planning more efficient.

1.3 Learning from Failures to Expand Capabilities

Neither slow reasoning nor fast execution can guarantee reliable robot behavior in the complex real world. Even with abstract world models and real-time, reusable capabilities, robots can still fail in situations outside their training distribution. Rather than treating these failures as endpoints, I explored how robots can use them to identify capability gaps and improve their models. In Fail2Progress [13], I developed a failure-driven adaptation framework that detects failures during real-world execution, triggers recovery through replanning, generates targeted data around the observed failure, and uses this data to refine the world model. This iterative loop of failure detection, recovery, and targeted model updates improves performance by **over 60%** in out-of-distribution environments, demonstrating the promise of refining robot models during deployment. Failure reasoning also benefits hierarchical systems built on foundation models. In collaborative work on routing deformable linear objects such as cables [14], a vision-language model plans over reinforcement-learned skills, detects execution failures such as unsuccessful clip insertions, and triggers a recovery skill before replanning, achieving 92% success across long-horizon routing scenarios.



Figure 4: When unexpected failures reveal gaps in existing robot capabilities, the robot can diagnose the failure, collect targeted data, and update its models and policies. This feedback loop enables robots to improve their capabilities through experience.

My work reframes failure as a trigger for capability expansion: when a real-time capability is insufficient, the robot can use reasoning to understand what went wrong, generate targeted experience, and update the capability so that similar failures become less likely in the future. This provides the feedback loop in my broader research program: deployment failures can trigger reasoning that creates future reusable capabilities.

2 Future Directions and Long-term Vision

My past work has explored how robots can build structured representations for reasoning, make reasoning more efficient, distill its solutions into real-time capabilities, and learn from failures during deployment. However, continually expanding these capabilities raises new challenges: deciding what knowledge to retain in memory, reasoning reliably when physical outcomes are uncertain, determining which capabilities can transfer across different robot embodiments, and adapting generalized capabilities to individual users. My future research will address these challenges through four complementary directions: structured robot memory, uncertainty-aware reasoning, generalizing capabilities across embodiments, and specializing capabilities to individual users.

Structured Robot Memory. Robots operating in partially observable environments must maintain knowledge of objects beyond their current observations. In my prior work [15], I combined learned world models with video tracking to reason about unobserved objects. However, retaining information about every tracked object can lead to unnecessary memory overhead as tasks and environments evolve. I will therefore investigate learned, task-aware memory gates that decide which object information to preserve, update, or discard based on its expected utility for future reasoning and planning. These gates will use current observations, existing memory, and task context to manage memory over long horizons. I will train these gates to preserve information that improves future planning while avoiding unnecessary memory and computation. The goal is to enable robots to accumulate compact, decision-relevant physical knowledge that improves long-horizon reasoning without requiring them to retain their entire interaction history.

Uncertainty-aware Reasoning. Physical interactions are inherently uncertain: contact, friction, and object dynamics can vary across environments, making the outcomes of robot actions difficult to predict reliably. In my prior work [16], we learned multimodal probabilistic models of manipulation effects in simulation, enabling robots to predict distributions over possible object pose changes rather than single-point estimates. Yet these

predicted distributions may not accurately capture real-world interactions because simulation and reality differ. I will therefore investigate how robots can adapt probabilistic world models using real-world interaction data while preserving useful knowledge learned in simulation. In particular, I will explore selectively updating the predicted outcome distributions of the probabilistic decoder, including their mixture weights, means, and covariances, without necessarily modifying the learned representations and latent dynamics. The goal is to enable robots to maintain accurate predictions of uncertain physical outcomes with limited real-world experience, supporting more reliable planning and decision-making.

Generalizing Capabilities Across Embodiments. A key challenge in building general-purpose robots is enabling capabilities acquired by one robot to transfer to others with different physical embodiments. My prior work has demonstrated that slow reasoning can provide solutions across multiple hardware platforms (two fixed-base manipulators [6,11], four mobile manipulators [1,8,10,13], and a surgical robot [17]), given appropriate perception and kinematic models, but its computational cost makes it impractical for real-time interaction. Building on DATAFARM [11], I envision using reasoning to generate training data across diverse robot platforms and fine-tune downstream VLAs. However, generating data across embodiments alone does not guarantee that models will learn transferable physical knowledge rather than memorize embodiment-specific behaviors. My prior work has explored visual reasoning across diverse embodiments through RoboVista [18] and demonstrated zero-shot cross-embodiment transfer in VLAs through LAP [19]. Building on these complementary foundations, I aim to investigate how slow reasoning across embodiments can provide supervision for learning transferable physical knowledge, beyond the transfer capabilities demonstrated by existing models. In particular, I will study how models can learn which aspects of a solution generalize across embodiments and which require adaptation to embodiment-specific constraints, such as kinematics and action spaces. Ultimately, I seek to turn reasoning across embodiments into real-time capabilities that transfer across robots, reducing the need to solve similar physical problems from scratch on each new robot.

Specializing Capabilities to Users. Even general-purpose robots must adapt to individual preferences and habits to provide meaningful value in everyday lives. I envision an off-the-shelf personalization layer built on top of generalized robot policies, allowing robots to adapt their behavior to individual users without retraining their entire policies or compromising their generalization capabilities. Building on Homeostasis at Home, a project I am currently pursuing with my mentee Jingru, I will investigate how robots can represent persistent user preferences, infer when those preferences apply, and adapt their behavior as new feedback becomes available. A central challenge is to learn user-specific preference representations from everyday observations and sparse feedback, and use them to adapt generalized policies without compromising their transferable capabilities. Ultimately, I aim to enable robots to reuse generalized capabilities while tailoring their execution to individual users, allowing them to integrate naturally into diverse, personalized environments.

Long-Term Vision. My long-term vision is to build robots that continually expand what they can accomplish in the physical world. Rather than being limited to tasks and environments encountered during training, these robots will reason about unfamiliar physical tasks, transform their solutions into fast, reusable capabilities, and continually improve through real-world experience. As robots accumulate reusable capabilities, they will be able to handle increasingly diverse tasks with less reliance on costly reasoning, while deployment failures will reveal when further reasoning is needed to expand their capabilities. These advances will enable robots to adapt to the changing needs of people in homes and care facilities, respond to evolving demands in industrial environments, and tackle unfamiliar challenges in hazardous settings such as disaster response. By transferring capabilities across tasks, environments, embodiments, and users, robots will become increasingly versatile and significantly expand the range of real-world tasks they can handle.

Funding and Collaboration. During my Ph.D., I had the opportunity to engage with DARPA-funded research activities and present my work directly to the DARPA TIAMAT program manager, which provided valuable exposure to federally funded robotics research. During my postdoc, I helped my advisor write a successful grant to Amazon Robotics and a proposal to Sony Research Awards that is currently under review. I am also helping prepare an NSF proposal, gaining experience in collaborative proposal writing. As a faculty member, I plan to seek research funding from federal agencies such as NSF, DARPA, and ONR. My research on reasoning and real-time, reusable robot capabilities aligns closely with programs such as the NSF Foundational Research in Robotics program, the DARPA TIAMAT program, and the ONR Young Investigator Program. I will also actively seek industry funding from NVIDIA, Google, Amazon Robotics, Sony, and Toyota Research Institute.

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